Cloud Architecture

Listing

- Monolithic
- Microservices
- Caching
- Authentication
- Database
- Ratelimit
- Containerization
- Queue
- CI/CD Pipeline
- Authorization

Grouping
- Backend
  - Database
  - Caching
  - Authentication
  - Authorization
  - Ratelimit
  - Queue

- Designs
  - Monolithic
  - Microservices

- Infrastructure
  - CI/CD Pipeline
  - Containerization

Questions

1. When should the startups different architectures be used?
2. What should be implemented in these architecture?
3. How should it be implemented?
4. Why is this architecture the best solution for implementation?

Sources:
- What is Cloud Architecture: https://www.ibm.com/think/topics/cloud-architecture
- Benefits and Components: https://cloud.google.com/learn/what-is-cloud-architecture

- Types and Designs of Cloud Architecture: https://www.cloudexmachina.io/blog/cloud-architecture

Topic
- Cloud Architecture at a Software Startup Company
- Choosing Architecture with building Products

Research Questions
- How should software companies choose which cloud architecture to implement?

Thesis Statement

To balance scarce initial resources with long-term growth, software startups must examine their cloud architecture, not as a permanent choice, but as an evolution from a streamlined monolithic structure to a modular micro-services architecture, together with automated infrastructure and essential backend services.

Claim

Monolithic architecture speeds up early startup development, but switching to a micro-services environment will make sure it survives in the long run.

Topic Sentence

For a startup company, navigating through initial resources and long term growth requires a cloud architecture that scales seamlessly from a monolithic architecture to a micro-services ecosystem.